File permissions in Linux

Project description

Used Linux commands to review and update file and directory permissions in a project workspace so access better matched the organization’s authorization requirements.

```
researcher2@39103d016a2e:~$ ls
projects
researcher2@39103d016a2e:~$ cd projects
researcher2@39103d016a2e:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 25 21:58 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 25 22:39 ..
-rw--w---- 1 researcher2 research_team   46 Mar 25 21:58 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 25 21:58 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Mar 25 21:58 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar 25 21:58 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 21:58 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 21:58 project_t.txt
researcher2@39103d016a2e:~/projects$ chmod o-w project_k.txt
researcher2@39103d016a2e:~/projects$ chmod g-r project_m.txt
researcher2@39103d016a2e:~/projects$ chmod ug-w,ug+r .project_x.txt
researcher2@39103d016a2e:~/projects$ chmod g-x drafts
researcher2@39103d016a2e:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 25 21:58 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 25 22:39 ..
-r--r----- 1 researcher2 research_team   46 Mar 25 21:58 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Mar 25 21:58 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 21:58 project_k.txt
-rw------- 1 researcher2 research_team   46 Mar 25 21:58 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 21:58 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 21:58 project_t.txt
researcher2@39103d016a2e:~/projects$
```

Check file and directory details

I used `ls -la` to view the contents of the `projects` directory, including hidden files and detailed permission settings. This showed one hidden file (`.project_x.txt`), one directory (`drafts`), and several project files, along with their current ownership and permissions.

Describe the permissions string

Each permissions string is 10 characters long. The first character shows whether the item is a file or directory. The next nine characters show read, write, and execute permissions for the user, group, and others. For example, `-rw-rw-r--` means the file owner and group can read and write the file, while others can only read it.

Change file permissions

- I removed write access for others from `project_k.txt`. This changed the file from world-writable to a more secure setting:
  chmod o-w project_k.txt

- I updated permissions on `.project_x.txt` so that neither the user (researcher2) nor group (research_team) had write access, but did have read access. This changed the hidden file to read-only for the user and group:
  chmod ug-w,ug+r .project_x.txt

- I removed execute permission for the group from the `drafts` directory, preventing group members from accessing or traversing the directory, as execute permission is required to enter a directory and access its contents:
  chmod g-x drafts

Summary

I used `ls -la` to inspect existing permissions and `chmod` to reduce unnecessary access to files and a directory. These changes better aligned permissions with least-privilege principles and improved the overall workspace security.